Declaring Multiple Variables

Efficiency vs. Readability in Java Data Storage

1. The "Comma" Shortcut

If you have several variables of the **same type**, you don't need to repeat the type keyword. Simply use a comma.

STANDARD STYLE

```
int x = 5;
int y = 6;
int z = 10;
```

SHORT STYLE

```
int x = 5, y = 6, z = 10;
```

2. Chained Assignment

Java allows a "chain reaction" for assignment. This is useful when all variables need the exact same starting value.

```
int x, y, z;
x = y = z = 50;
```

The value 50 flows from right to left, filling every box.

3. The "Same Type" Rule

The biggest restriction is that you **cannot mix different types** on the same line using the comma shortcut.

INVALID

```
int age = 20, double price =
5.5;
```

VALID

```
int age = 20; double price =
5.5;
```

4. Practical Implementation

In real-world scenarios, this is often used for grouping related data points, like coordinates or test scores.

```java
// Grouping related variables
int math = 80, science = 75, english = 90;
int total = math + science + english;

System.out.println("Final Total: " + total);
```

5. Common Pitfalls

⚠️ **The Declaration Gap:**
If you write `int x, y, z = 50;` only **z** is given a value.
x and y are created but stay empty. Trying to use x + y will cause an error!

6. Best Practice Tip

💡 While the comma shortcut saves space, **Professional Developers** usually prefer writing one variable per line with a comment. It makes your code easier to read when your program grows from 10 lines to 1,000 lines.

Summary: Use commas for related variables of the same type. Use chain assignment (x=y=z) for shared values. Prioritise readability over "saving lines".